

SIMATS ENGINEERING

ASSESSMENT TOOL – I : CONCEPT MAPPING

Course Code	ECA14
Course Name	Embedded Systems
Course Outcome	CO1
Assessment Tool	Concept Mapping
Question No.	
Question Topic	Develop a comprehensive concept map that links Finite State Machine (FSM) concepts — including states, transitions, inputs, outputs, Moore and Mealy models, state minimisation, and reachability — and connect these elements to the formal design and verification of an embedded controller such as a traffic light system or vending machine.
Bloom's Taxonomy Level	BTL 4 / BTL 5

Evaluation Rubric (20 Marks)

Criteria	Weightage	Level 4	Level 3	Level 2	Level 1	Marks Awarded
Concept Identification	5	☐ 5	☐ 4	☐ 3	☐ 1–2	
Concept Linkages	5	☐ 5	☐ 4	☐ 3	☐ 1–2	
Organization & Structure	4	☐ 4	☐ 3	☐ 2	☐ 1	
Integration of Literature	4	☐ 4	☐ 3	☐ 2	☐ 1	
Visual Clarity & Presentation	2	☐ 2	☐ 1.5	☐ 1	☐ 0.5	
Total	20					/20

Code of Conduct:

I ____ Sangana Raviteja/ Regno.192312283 _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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ASSESSMENT TOOL – II : DESIGN TASK

Course Code	ECA14
Course Name	Embedded Systems
Course Outcome	CO1
Assessment Tool	Design Task
Question No.	
Question Title	Analyze the embedded software testing strategy for an automotive ECU that experienced a safety-critical race condition during field testing — two CAN messages arriving within 500 μ s caused a vehicle stall at 80 km/h. Examine which testing level (unit, integration, HIL, system) should have detected this defect, identify the coverage metric gap (MC/DC), calculate the cost of detecting this bug at the current phase vs. design review, and redesign the testing process to structurally prevent such escape defects.
Maximum Marks	100
Bloom's Taxonomy Level	BTL 5

Evaluation Rubric (100 Marks)

Criteria	Weightage	Level 4	Level 3	Level 2	Level 1	Marks Awarded
Understanding of Design	20	☐ 20	☐ 16	☐ 12	☐ 8	
Evaluation of Design	25	☐ 25	☐ 20	☐ 15	☐ 10	
Justification	20	☐ 20	☐ 16	☐ 12	☐ 8	
Suggestions for Improvement	20	☐ 20	☐ 16	☐ 12	☐ 8	
Written Communication	15	☐ 15	☐ 12	☐ 9	☐ 6	
Total	100					/100

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